Understanding Implementation of Shor's Algorithm in Quantum Computing

Adrian Vilchez
Department of Computer Science
University of California, Santa Cruz
March 2026

***Abstract*—Quantum computing is one of the most important emerging technologies. One of its crown jewels is Shor's algorithm, which factors large integers in polynomial time and threatens sytems such as RSA. This paper presents a development of quantum computing. Starting with foundational principles and going through the implementation of Shor's algorithm. It covers qubits, superposition, entanglement, the quantum Fourier transform, and quantum phase estimation before any analysis of Shor's algorithm and its experimental realization. The discussion concludes with the physical limitations and ethical implications of practical quantum computation. This is primarily intended for advanced undergraduate students in computer science, physics, and electrical engineering, but is remains open to those outside of these fields.**

Keywords— Quantum computing, Shor's algorithm, quantum Fourier transform, cryptography, quantum phase estimation

I. Introduction

The word "quantum" shows up in media, politics, news, but it is often misunderstood or overstated. Quantum computing, however, isn't marketing language or an obscurity. It is a well established rigorous branch of Computer Science and Electrical Engineering, with theoretical roots laying in the 1980s. At the core, it relies on key principles of quantum mechanics -superposition, entanglement, and interference- to perform certain computations that are impractical on classical hardware.

Few results capture the importance of a quantum computer as clearly as Shor's algorithm, first introduced by Peter Shor in 1994. The algorithm demonstrated that a quantum computer would be able to factor a larger Integer N in time that scales in polynomial time with the number of bits needed to represent N. In contrast, classical algorithms would need exponential time. This difference highlights precisely why systems such as RSA are threatened with the implementation of large-scale quantum computers. This possibility has driven intense global races in both quantum computing hardware development and post-quantum cryptography.

This paper presents a straightforward overview of quantum computing, starting from the basics and building up to advanced concepts such as the quantum Fourier transform and Quantum Phase Estimation, ultimately ending in the account for Shor's algorithm. Specifically, the 2001 experimental realization using nuclear magnetic resonance (NMR) by Vandersypen, Steffen, Breyta, Yannoni, Sherwood, and Chuang at IBM Almaden and Stanford University is used as the prime example. The paper concludes by situating these theoretical achievements within the physical limitations and ethical questions that currently constrain and complicate the field.

The work draws on several sources, including lecture material from IBM Qiskit Global Summer School, which explains and covers quantum phase estimation and Shor's algorithm in pedagogical depth; the undergraduate survey paper "Quantum Computers" by Adrian Vilchez [12]; and the Nature paper "Experimental Realization of Shor's Quantum Factoring Algorithm Using Nuclear Magnetic Resonance" by Vandersypen et al. (2001). Together, the sources givea mix of explanations, deep insight, and eventual world experimentation.

II. Classical Versus Quantum Information: Bits and Qubits

To understand quantum computing, it is best to understand how classical computing works. Classical systems are built on bits, either 0 or 1. Physically these states correspond to voltage levels in a transistor or magnetic orientation on a disk. At any time, the bit is only in one of these two states. A classical computer processes this information by manipulating strings of bits according to the rules of Boolean logic. The power of modern classical computers comes from the ability to make billions of these switches onto a single silicon chip and change their states at billions of cycles per second [3], [12].

Quantum computers use quantum bits, qubits, instead of bits. This is a two-level quantum system—for example, the spin of an electron, the polarization state of a photon, or the energy levels of a trapped ion or a superconducting circuit. A qubit can exist in a combination of both of the states, known as superposition, and written as $|0\rangle$ and $|1\rangle$. Mathematically, the state of the qubit is written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex probability amplitudes which satisfy $|\alpha|^2 + |\beta|^2 = 1$. The quantity $|\alpha|^2$ gives the probability of finding the qubit in state $|0\rangle$ upon measurement, and $|\beta|^2$ gives the probability of finding it in state $|1\rangle$. Before measurement, the qubit is genuinely in both states simultaneously. This is the fundamental difference between classical bits and quantum bits. [1], [3].

The difference becomes much more significant as we look at multiple qubits. In a classical system a register of n bits can store exactly one of 2^n possible values at any given time. A quantum system however with a quantum register of n qubits, can exist in a superposition of all 2^n basis states simultaneously. At 32 qubits, this encodes 2^{32}, or over four billion, values. At 300 qubits, the register can represent more distinct states than there are molecules in the observable universe [12]. This exponential scaling is the source of quantum computing's theoretical power—and its exploitation is exactly what Shor's algorithm requires as will be demonstrated.

The physical implementation of qubits presents many engineering challenges. Current platforms include trapped ions, photons, artificial atoms realized in superconducting circuits, and electrons in silicon. Each of these systems exploits the quantum properties described, and each comes with its own error rates, coherence times, and fabrication requirements. Materials scientist Andrea Morello in [12], states that the production of silicon qubits requires highly isotopically purified silicon-28, since silicon-29 introduces nuclear spin noise that causes decoherence. Decoherence here is the gradual loss of quantum information due to unwanted interactions with the environment. These material requirements show the gap between the elegant mathematics of qubits and the terribly messy reality of building them.

III. Superposition, Measurement, and Quantum Gates

Quantum computers manipulate the sates of qubits using quantum gates. This is the analogue to logical gates in classical computers. Quantum gates are represented by unitary matrices, these are those which U which satisfy $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose and I is the identity matrix. This condition ensures that quantum evolution is reversible and probability-preserving. In other words, the total probability of all measurement outcomes always sums to one [3].

One of the most important quantum gates is the Hadamard gate, denoted by H. Applied to a qubit in the state $|0\rangle$, it produces the superposition

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

an equal superposition with both amplitudes equal to $1/\sqrt{2}$. Applied to $|1\rangle$, it produces

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The Hadamard gate is the quantum analogue of a fair coin flip, this places the qubit in an equal mixture of zero and one. As a Unitary matrix, this application is reversible, where application of the matrix twice returns the qubit to its original state. This is the fundamental feature that distinguishes a quantum gate from a classical gate. [9].

Another family of gates are the phase rotation gates, often noted as R_k, which applies a phase of $e^{2\pi i/2^k}$ to the $|1\rangle$ component of a qubit while leaving the $|0\rangle$ component completely unchanged. The gates here are critical in the construction of the Quantum Fourier transform, as discussed in Section IV. Together with the Hadamard gate, H and the controlled-NOT (CNOT), which flips the qubit if and only if a control qubit is in state $|1\rangle$, forms gate set sufficient to implement any quantum algorithm [3], [9].

Additionally, the superposition principle has a subtlety worth emphasizing: quantum states cannot be copied. This no-cloning theorem, a direct consequence of the linearity of quantum mechanics, means that one cannot simply read off the state of a qubit without disturbing it. After measuring a qubit of the state $\alpha|0\rangle + \beta|1\rangle$, either $|0\rangle$ with probability $|\alpha|^2$ or $|1\rangle$ with probability $|\beta|^2$ will be returned. In other words, measurement causes the collapse of superposition. The act of measurement is also irreversible and when it destroys the superposition. This feature is the tension in quantum algorithm design. The goal to arrange the quantum circuit so that when measurement does occur, the desired answer is given with high probability. Achieving this requires interference, which is the principle in quantum mechanics that can be constructive and destructive, a combination of probability amplitudes. This quality is the third key quantum phenomenon exploited by algorithms such as Shor's [3], [9].

IV. Quantum Entanglement

Superpostion by itself isn't the only highly important quality of quantum computing. Entanglement, the correlation between quantum systems that really has no analogue in classical computation, is essential to the power of quantum computing. If two qubits are entangled then their joint state cannot be written as a product of their individual qubit states. An example of this is the four Bell states, where the simplest is,

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

This state, neither qubit has a definite value before measuring, but on measurement, the first qubit immediately determines the state of the second, regardless of the distance between them. [9].

This correlation is stronger than any classical correlation. Classical correlation can be explained by pre-existing information. The analogy of two cards dealt face down, one black and other the red, where learning one reveals the other. Quantum entanglement, on the other hand, violates Bell inequalities the prediction of entangled quantum states cannot be reproduced by any hidden variable theory. This is confirmed experimentally to extraordinary precision from Aspect's experiments in the 1980s. The Transcript presentation from the Summer School explains the Bell inequality framework in detail, noting that maximally entangled states achieve the maximum possible violation, with CHSH parameter value $2\sqrt{2}$ rather than the classical bound of 2 [9].

In quantum computation, entanglement is useful for several ideas. The first, is that it allows the creation of correlation between qubits which then directly enables parallel computation. The second, it is the resource exploited in quantum teleportation and superdense coding. The third, in Shor's algorithm, the entanglement created by modular exponentiation is what causes the quantum Fourier transform to show the period of a function. Without this principle, quantum computers would practically be at the same usage of classical computers [3].

Entanglement also creates challenges. For example, when qubits are entangled, measuring collapses the join state and determines the state of the other. This phenomena initially seems to threaten special relativity, however, it stands that faster-than-light communication is not violated. Still, the physical mechanism underlyiung remains the subject of ongoing investigation. [12].

V. The Quantum Fourier Transform

The quantum Fourier transform (QFT) is the quantum analogue of the discrete Fourier transform (DFT) and is a key subroutine in both quantum phase estimation and Shor's algorithm. classical DFT maps a vector of N complex numbers to another vector of N complex numbers by using the formula

$$\tilde{x}_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{2\pi ijk/N}.$$

The best classical algorithm for this computation, the fast Fourier transform, requires the use of $O(N \log N)$ operations. The QFT achieves the same transformation on an n-qubit quantum state—where $N = 2^n$—using only $O(n^2)$ gates. The speedup here is exponential, but with an important subtly. The QFT cannot be used to read off all the Fourier coefficients directly, since measurement collapses the state [3], [9].

The power of QFT in quantum algorithm is not in the extraction of the all the coefficients, but instead in using the Fourier-basis as a representation to make certain information accessible via measurement. In specific, if a quantum state has the periodicity r in the computational basis, the QFT will map it to the state whose support is all focused on multiples of $2^n/r$ in the Fourier basis. The measurement will then return one of the values with high probability, allowing r to be recovered by classical post-processing. Additionally, if it is missed, the simulation must simply be run with sufficient trials. This is the most essential mechanism exploited by Shor's algorithm [3], [9].

The circuit for the n-qubit QFT is constructed from Hadamard gates and controlled phase rotations. The action on a computational basis state $|x\rangle = |x_{n-1} \dots x_1 x_0\rangle$, where x_k are the binary digits of x, can be written as a tensor product of single-qubit states:

$$\text{QFT}\,|x\rangle = \bigotimes_{k=1}^{n} \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \cdot 0.x_{n-k} \dots x_0} |1\rangle\right),$$

where $0.x_{n-k} \dots x_0$ denotes a binary fraction. This product will reveal that the QFT maps a qubit to a superposition with a phase that is dependent on the binary representation of x, and the circuit can be implemented by applying a Hadamard to the first qubit. Then, a series of controlled phase rotations from subsequent qubits, followed by a swap network to reverse the qubit ordering [3], [9].

The Transcript provides a careful pedagogical development of this circuit, including the observation that the output ordering is reversed relative to the natural convention, a technical point often handled by appending swap gates or by keeping track of the reversal in subsequent processing. The Transcript instructor emphasizes that the QFT circuit is conceptually simple once the product-form expression is understood, and that the Hadamard gate is the special case of a phase rotation with $k = 1$ [9].

VI. Quantum Phase Estimation

Quantum phase estimation (QPE) is another fundamental quantum subroutine that uses the QFT to extract the eigenvalue phase of a unitary operator. The importance of the QPE extends even beyond Shor's algorithm. QPE is the basis of quantum simulation algorithms, quantum chemistry, and the HHL algorithm for linear systems. Understanding the QPE essential for any serious study of quantum algorithm. [4], [9].

QPE solves the following problem, given a unitary operator U and an eigenstate
$|\psi\rangle$ satisfying

$$U\,|\psi\rangle = e^{i\theta_\psi}\,|\psi\rangle\,,$$

estimates the eigenvalue phase θ_ψ. The naive approach—measuring a global phase—is impossible in general because global phases do not affect measurement probabilities. QPE converts the phase into amplitude information that can be measured, using a circuit that is a direct precursor to Shor's algorithm [9].

The single-qubit version of QPE can illustrate the core idea. Begin with the ancilla qubit in $|0\rangle$, apply a Hadamard to produce

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

then apply the controlled-U gate—which applies U to $|\psi\rangle$ if and only if the ancilla is $|1\rangle$. Finally, apply a second Hadamard. The resulting state before measurement is,

$$\frac{1}{2}\left[(1 + e^{i\theta_\psi})\,|0\rangle + (1 - e^{i\theta_\psi})\,|1\rangle\right]|\psi\rangle\,.$$

The probabilities of measuring $|0\rangle$ and $|1\rangle$ in the ancilla are $|1 + e^{i\theta_\psi}|^2/4$ and $|1 - e^{i\theta_\psi}|^2/4$ respectively, which differ by an amount that encodes θ_ψ. For a small θ_ψ, however, these probabilities are almost equal and many measurement trails are then required for the extraction with good precision. θ_ψ [9].

The n-qubit generalization addresses this exact precision problem. Prepare the n ancilla qubits in the $|0\rangle$ state, then apply Hadamards, then apply controlled-U^{2^k} gates for $k = 0, 1, \dots, n-1$, where each gate is controlled by the k-th ancilla qubit.

After the controlled operations, the n ancilla qubits ends up in a state that is equivalent to applying the QFT to $|2^n\theta_\psi/2\pi\rangle$. By then applying the inverse QFT and measuring, a result of n-bit estimate of $2^n\theta_\psi/2\pi$ with high probability is obtained. The key point is the factor of 2^n: increasing the number of ancilla qubits will directly improve the precision of the estimate. For example, using $n = 10$ qubits yields about 1024 times finer resolution than a single-qubit estimate. As found in the Transcript, adding more qubits to the estimation register leads to a correspondingly sharper determination of the phase [9].

This perspective also makes the connection between QPE and the QFT more concrete. The sequence of controlled operations effectively builds the QFT of the phase information into the ancilla register, and the inverse QFT converts this information back into a form that can be read out through measurement. This is why the QFT appears so frequently in quantum algorithms: many problems, including period finding, naturally produce structure that is most clearly revealed in the Fourier domain [3], [9].

VII. Shor's Algorithm: Factoring to Period Finding

Shor's algorithm, first introduced by Peter Shor in 1994 and later formalized in his 1997 SIAM Journal paper, shows that integer factorization can be performed in polynomial time on a quantum computer [7]. In contrast, the best known classical method—the general number field sieve—runs in time on the order of $\exp(O(n^{1/3}))$, where n is the number of bits in the integer being factored. For numbers of the size used in modern cryptography, such as 2048-bit RSA keys, this makes classical factoring effectively impossible in practice. Shor's algorithm, however, completes the same task in $O(n^3)$ quantum operations, representing an exponential improvement and posing a direct threat to widely used cryptographic systems such as RSA, Diffie-Hellman, and elliptic curve cryptography [3], [11].

The main idea Shor conveys is that the algorithm works by reducing factoring to a different type of problem, finding the period of a modular exponential function. Given an integer N, select a random integer A such that $1 < A < N$ and $\gcd(A, N) = 1$, in other words, let the two integers be co-prime. Study the function $f(x) = A^x \bmod N$. This function is periodic, with period r defined as the smallest positive integer such that $A^r \equiv 1 \pmod N$. If r is even and $A^{r/2} \not\equiv -1 \pmod N$, then computing $\gcd(A^{r/2} \pm 1, N)$ yields a non-trivial factor of N. For semiprime N, a randomly chosen A satisfies these conditions with probability greater than $1/2$, so only a small number of trials are usually needed [3], [11].

The Transcript illustrates this idea with a simple example using $N = 15$. Choosing $A = 13$, one computes $13^x \bmod 15$ for $x = 0, 1, 2, 3, 4$, obtaining the sequence $1, 13, 4, 7, 1$, which reveals a period of $r = 4$. Because r is even, one computes $A^{r/2} \bmod N = 13^2 \bmod 15 = 4$. From this fact, $\gcd(4 + 1, 15) = 5$ and $\gcd(4 - 1, 15) = 3$, successfully recovering the factors of 15 [9]. This same example also appears in the experimental work of Vandersypen et al. [11], where related choices of A were then used to demonstrate the algorithm physically.

On first glance, a quantum advantage does not seem obvious, however, it is present at the period-finding step. Classically, finding the period of $A^x \bmod N$ requires evaluating the function for many values of x, which quickly becomes impractical as N grows. The quantum approach avoids this by using superposition through an evaluation of $f(x)$ for many inputs.

Prepare the state $\sum_x |x\rangle |0\rangle$, apply modular exponentiation to obtain $\sum_x |x\rangle |A^x \bmod N\rangle$, and then apply the inverse QFT to the first register. Measuring this register produces a value close to $c \cdot 2^n/r$ for some integer c, from which the period r can be recovered using the continued fractions algorithm on a classical computer. While modular exponentiation still remains the most complex part of the circuit, it can still be implemented more efficiently using quantum versions of standard arithmetic techniques rather than classical ones. [3], [11].

As described in the Transcript, the overall circuit follows a clear structure. Two registers are prepared: the first with n qubits initialized to $|0\rangle^n$ (chosen so that $2^n > N^2$), and the second with $m = \lceil \log_2 N \rceil$ qubits initialized to $|0 \cdots 01\rangle$. Applying the set of Hadamard gates to the first register creates an equal superposition over all inputs. The modular exponentiation operation then entangles the two registers. The second register may be measured (though this is not strictly necessary), and the inverse QFT is applied to the first register. A final measurement yields a value related to $2^n/r$, from which the period can be determined [9]. Conceptually, this structure closely parallels the quantum phase estimation framework discussed earlier, with $f(x) = A^x \bmod N$ playing the role of the unitary whose phase encodes the period.

VIII. Experimental Realization: The NMR Implementation

The first realization of Shor's algorithm in an experiment was reported in 2001 by Vandersypen, Steffen, Breyta, Yannoni, Sherwood, and Chuang at IBM's Almaden Research Center who was in collaboration with Stanford University. In this experiment, they factored $N = 15$ using a custom-designed molecule as a quantum processor. In specific they used seven nuclear spin-$\frac{1}{2}$ nuclei acting as qubits [11].

The molecule—a perfluorobutadienyl iron complex containing two ^{13}C-labeled carbon atoms—provided five ^{19}F spins and two ^{13}C spins. These seven spins were well separated in resonance frequency, which made it possible to address them individually using radio-frequency (RF) pulses. The interactions between spins were governed by J-coupling, a scalar interaction described by the Hamiltonian

$$H_J = \sum_{i<j} 2\pi\hbar J_{ij} I_{zi} I_{zj}.$$

Quantum gates were also implemented as sequences of roughly 300 carefully timed RF pulses, interleaved with periods of free evolution. The pulses lasted between 0.22 and 2 milliseconds each, and the full pulse sequence for Shor's algorithm took about a mere 720 milliseconds to execute [11].

The experiment was carried out for two representative values of A: $A = 11$, which led to a relatively simple circuit, and $A = 7$, which required a more involved sequence of operations. In the case where $A = 11$, modular exponentiation simplifies because $11^2 \bmod 15 = 1$, so only a small number of controlled operations were necessary. For $A = 7$, however, $7^2 \bmod 15 = 4 \neq 1$, so the additional controlled multiplications must be implemented. The inverse QFT, following the construction of Coppersmith 1994 [2], required approximately 120 milliseconds [11].

One of the main experimental challenges here was state preparation. At room temperature, NMR systems begin in a thermal equilibrium state described by

$$\rho_{\text{th}} = \frac{e^{-H_0/k_B T}}{2^7},$$

where $k_B T \gg \hbar\omega_i$. In this regime, each spin is almost equally as likely to be in $|0\rangle$ or $|1\rangle$, resulting in a highly mixed state. To address this, Vandersypen et al. used temporal averaging to construct an effective pure state ρ_1. This required 36 individual experiments (organized as 4 sets of 9), whose results were combined. Although the system remained physically mixed, the resulting signal behaved as if the system had been initialized in $|0000001\rangle$, allowing the algorithm to proceed as intended [11].

Measurement was performed using NMR spectroscopy. After the computation, the states of the first three qubits were inferred from their spectra. For the simpler case ($A = 11$), the results indicated that qubits 1 and 2 were in $|0\rangle$, while qubit 3 was in an equal mixture of $|0\rangle$ and $|1\rangle$, corresponding to a superposition of $|000\rangle$ and $|100\rangle$. The outcome implies a period $r = 2$, from which the factors of 15 can be recovered. For the more complex case ($A = 7$), the observed mixture of $|000\rangle$, $|010\rangle$, $|100\rangle$, and $|110\rangle$ corresponds to a period $r = 4$, again, which yields the correct factors [11].

Decoherence was identified as the primary source of error in the experiment. The authors modeled this using a parameter-free approach based on the Bloch equations, incorporating both amplitude damping (with time constant T_1) and phase damping (with time constant T_2). By applying this model independently to each spin and combining the results, they were able to reproduce the main features of the experimental spectra with good accuracy. Notably, this analysis showed that decoherence, rather than imperfect control, was the dominant limitation, suggesting that the pulse sequences themselves were implemented with high fidelity [11].

The authors emphasized that this demonstration does not scale to larger problem sizes. In particular, the temporal averaging method requires a number of experiments that grows exponentially with the number of qubits, making it impractical for larger systems. As a result, NMR is not considered a scalable platform for quantum computing. Even so, the experiment remains important as an early proof of concept, demonstrating that Shor's algorithm can be implemented in a real physical system and helping establish experimental techniques that continue to influence modern quantum hardware development [11].

IX. Physical Challenges in Quantum Hardware

The experimental success of Vandersypen et al. notwithstanding, the practical realization of a large-scale, fault-tolerant quantum computer capable of running Shor's algorithm against cryptographically relevant key sizes remains a formidable engineering challenge. The obstacles fall into three interconnected categories: error correction, hardware scalability, and physical operating conditions.

Quantum error correction (QEC) is fundamentally different from classical error correction, and its existence was not obvious at the outset. As noted by [3], the problem seems paradoxical: error correction requires measuring the state of a system, but measurement collapses quantum superpositions, potentially destroying the very information one seeks to protect. The resolution, first demonstrated independently by Peter Shor and Andrew Steane in 1995, is to encode logical qubits in entangled states of multiple physical qubits in such a way that errors can be detected through syndrome measurements that reveal information about the error without revealing information about the encoded logical state. The Shor code uses nine physical qubits per logical qubit; the Steane code uses seven. Subsequent developments—including topological codes such as the surface code—have produced more efficient encodings, but all require significant overhead: current estimates suggest that breaking RSA-2048 with Shor's algorithm would require millions of physical qubits, compared to the few thousand logical qubits the algorithm itself requires, once error correction overhead is accounted for [3], [12].

Hardware scalability presents its own difficulties. As described in the [12] paper, Morello has noted that increasing the speed of a quantum computer generally increases its error rate—a direct consequence of the trade-off between gate speed and the precision required for coherent manipulation. Physical quantum computers must operate under extreme conditions: superconducting qubit systems, currently the leading platform in terms of qubit count, require temperatures below 15 millikelvin, colder than deep space, and the dilution refrigerators required to achieve these temperatures occupy significant physical space. Trapped ion systems offer better coherence times but slower gate speeds and present different scalability challenges related to ion crystal control. The silicon spin qubit platform studied by Morello's group offers the attractive prospect of leveraging semiconductor fabrication technology, but requires isotopically purified silicon-28 to eliminate decoherence from ^{29}Si nuclear spins [12].

The concept of quantum volume—a benchmark introduced by IBM that accounts for qubit number, connectivity, gate fidelity, and other hardware characteristics—provides a more holistic measure of quantum computer capability than qubit count alone. Progress in quantum volume has been rapid, with IBM, Google, IonQ, and other companies announcing regular improvements. Google's 2019 claim of quantum supremacy—performing a sampling task in 200 seconds that they estimated would take a classical supercomputer 10,000 years—marked a widely publicized milestone, though the classical benchmark has subsequently been revised downward by improved classical algorithms. The field's honest assessment is that fault-tolerant, cryptographically relevant quantum computation remains years to decades away, with the precise timeline depending on progress in multiple concurrent engineering challenges [12].

X. Ethical and Security Implications

The potential of quantum computers to break RSA encryption is not merely a theoretical curiosity—it has immediate and serious implications for national security, commercial infrastructure, and individual privacy. As [3] observes, any computer capable of efficiently finding the period of $A^x \bmod NN$ would pose an enormous threat to both military and commercial communications, since RSA encryption underpins the secure transmission of financial transactions, health records, classified government communications, and the authentication systems that underlie virtually all internet security.

[12] elaborates on the scale of the risk, noting projections—attributed to intelligence analysis—that a sufficiently advanced quantum computer might break 2048-bit RSA encryption in eight hours or less. The concern is compounded by the adversarial dimension: a nation or organization that achieves large-scale quantum computing capability before others could retroactively decrypt communications intercepted and stored today in anticipation of future cryptanalytic capability—a strategy sometimes called "harvest now, decrypt later." The urgency of this scenario has motivated the U.S. National Institute of Standards and Technology (NIST) to conduct a multi-year post-quantum cryptography standardization process, which in 2024 published its first set of post-quantum cryptographic standards based on lattice problems and hash functions believed to be resistant to quantum attacks.

Beyond cryptography, quantum computing raises questions about equitable access and dual-use potential. The enormous financial and technical resources required to build large-scale quantum computers mean that capability will likely be concentrated initially among a small number of nation-states and well-capitalized technology corporations. This concentration creates asymmetric risks: those who possess quantum computers gain advantages not only in cryptanalysis but potentially in drug discovery, materials science, optimization, and financial modeling. The societal implications of such concentrated capability—and the governance frameworks needed to manage them responsibly—are questions that scientists, policymakers, and ethicists are only beginning to grapple with in a serious way [10], [12].

It bears emphasizing that the same quantum computing capabilities that threaten existing cryptography also enable quantum key distribution (QKD) and other quantum cryptographic protocols that are provably secure against any computational adversary, including quantum computers. The transition to post-quantum and quantum-native security architectures is a major ongoing effort in the cryptographic community, and its success will depend partly on how quickly quantum hardware capabilities mature relative to the deployment of quantum-resistant alternatives [3].

XI. Discussion

Quantum computing introduces both technical breakthroughs and societal risks. The ability to break RSA encryption raises immediate cybersecurity concerns, while advances in simulation and optimization promise benefits in science and engineering.

This paper contributes a unified explanation of how foundational quantum principles connect directly to real-world consequences. It confirms existing research while highlighting the gap between theoretical capability and physical realization.

XII. Conclusion

This paper traced the development of quantum computation from its principles: qubits, superposition, entanglement, and quantum gates. Through the mathematical work of the Quantum Fourier Transform and the Quantum Phase Estimation to the theory and experimental implementation of Shor's algorithm, the demonstration is now complete. The most critical feature surveyed here is the reduction of integer factorization to a periodicity optimization problem, which can be solved in polynomial time on a quantum computer through the constructive interference done via the QFT.

The experimental work of Vandersypen et al. (2001) demonstrates that this reduction is not just mathematical, but a physical possibility and reality. Using nuclear magnetic resonance on a seven-qubit molecule, factorization of N = 15 was successful and confirmed the claim of Shor's algorithm under a realistic decoherence condition. The parameter-free decoherence model, which predicted the spectra without free parameters, is a representation of contributions to the practicality of quantum systems and illustrates the work between theory and experiment which characterizes the field.

The path from this proof-of-concept to relavent quantum computers remain far and demanding. Error correction exists as overhead; coherence times, operating temperatures, and the fabrication precision all are challenges that will require major discoveries for the advancement of the field. Still, the case for quantum computing's power is not a doubt, and the pace of the progress has been as accelerated and accelerating since the NMR demonstration. The Qiskit Summer School lectures captured reflect a field in which the tools for building and programming quantum computers are now accessible to advanced undergraduates, a democratization of quantum computing education that would have been unimaginable at the time of Shor's original paper.

The ethical stakes of this technology are huge. The ability to break public-key RSA would have consequences from the exposure of private communication to the destabilization of financial systems and national security. These risks make the transition to post-quantum standards not only a matter of interest but an urget priority. This underlines the responsibility of the quantum computing community -to pursue this science with care and clarity for its potential and risks and have serious engagement with ethical questions, including its governance, as the field will surely develop.

Best said by the Transcript instructor, after learning the quantum Fourier transform, quantum phase estimation, and Shor's algorithm in rapid development over a summer school: "You've gone from not just learning about these quantum algorithms, but to now zooming out and using these algorithms as black boxes in bigger things." The capacity to zoom out—to see how fundamental quantum phenomena chain together into algorithms with world-altering implications—is both an intellectual reward and the primary reason it demands out utmost careful attention.

REFERENCES

[1] G. P. Berman, G. D. Doolen, R. Mainieri, and V. I. Tsifrinovich, *Introduction to Quantum Computers*. Singapore: World Scientific, 1998.

[2] D. Coppersmith, "An approximate Fourier transform useful in quantum factoring," IBM Research Report RC19642, 1994.

[3] N. D. Mermin, *Quantum Computer Science: An Introduction*. Cambridge, U.K.: Cambridge University Press, 2007.

[4] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge, U.K.: Cambridge University Press, 2000.

[5] R. Rietsche, C. Dremel, S. Bosch, L. Steinacker, M. Meckel, and J.-M. Leimeister, "Quantum computing," *Electronic Markets*, vol. 32, no. 3, 2022. Available: https://doi.org/10.1007/s12525-022-00570-y

[6] P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in *Proc. 35th Annu. Symp. Foundations of Computer Science (FOCS)*, 1994, pp. 124–134.

[7] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM J. Comput.*, vol. 26, no. 5, pp. 1484–1509, 1997.

[8] S. P. Suprun and A. P. Suprun, *Computers: Classical, Quantum and Others*. Bentham Science Publishers, 2011.

[9] IBM Research, "IBM Qiskit Global Summer School: Quantum phase estimation and Shor's algorithm," Lecture recording and transcript, 2020–2021.

[10] P. B. Upama, "Evolution of quantum computing: A systematic survey on the use of quantum tools," in *Proc. Int. Conf.*, 2022, pp. 513–527.

[11] L. M. K. Vandersypen, M. Steffen, G. Breyta, C. S. Yannoni, M. H. Sherwood, and I. L. Chuang, "Experimental realization of Shor's quantum factoring algorithm using nuclear magnetic resonance," *Nature*, vol. 414, pp. 883–887, 2001.

[12] A. Vilchez, "Quantum Computers," Writing 2: Rhetoric and Inquiry, Univ. of California, 2023.